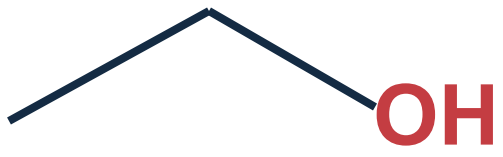
Molecules have connectivity and a 3D shape

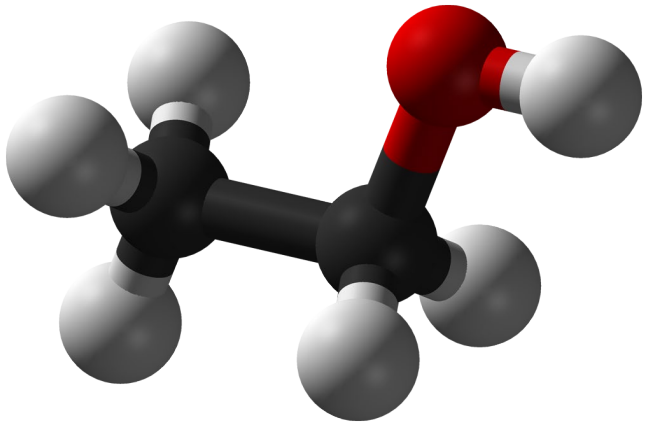
Ethanol is a small teaching example: C_2H_6O

2D structure



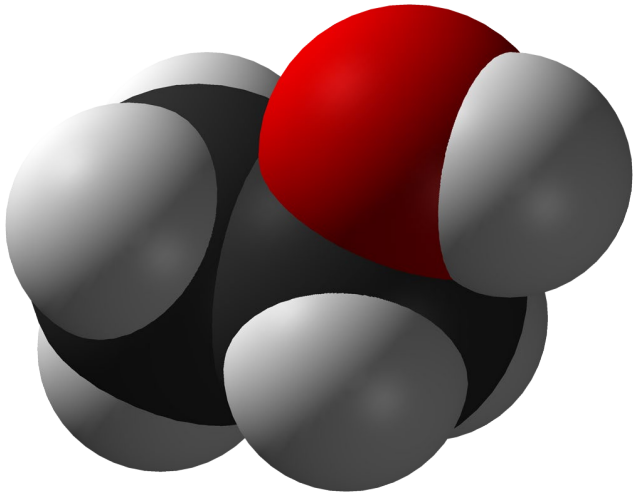
Which atoms connect?

Ball-and-stick model



Bond geometry becomes visible

Space-filling model

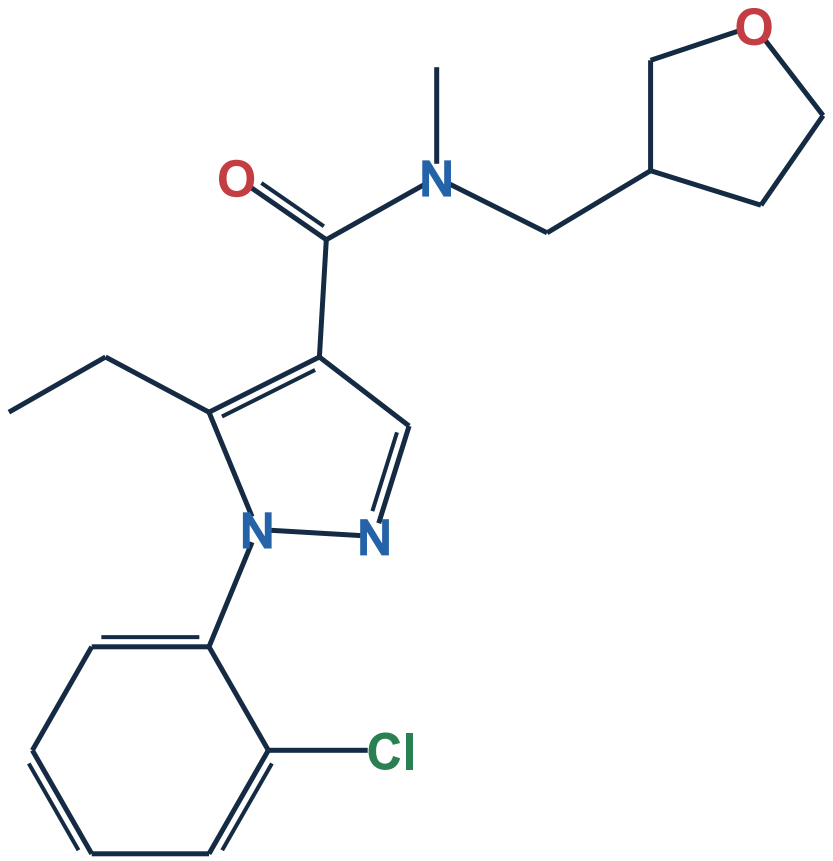


Atoms occupy space

Carbon and many hydrogens are implicit in the 2D drawing. Grey = carbon, white = hydrogen, red = oxygen.

Reading a real phase1 molecule

CatalogID Z1840178734, taken from the local structure file



Corners and ends represent carbon

Hydrogens on carbon are usually omitted.

Letters identify other elements

N = nitrogen O = oxygen Cl = chlorine

Lines show covalent bonds

Parallel lines indicate a double bond.
Closed paths form molecular rings.

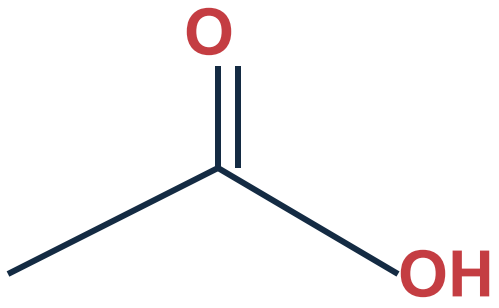
A flat drawing records connectivity. It does not show a measured binding pose.

SMILES serialises a molecular graph

A character string records atoms and bonds in a traversal of the graph

CC(=O)O

Acetic acid



C Carbon atom

(=O) Branch with a double bond to oxygen

c1ccccc1

Aromatic carbons; matching “1” closes a ring

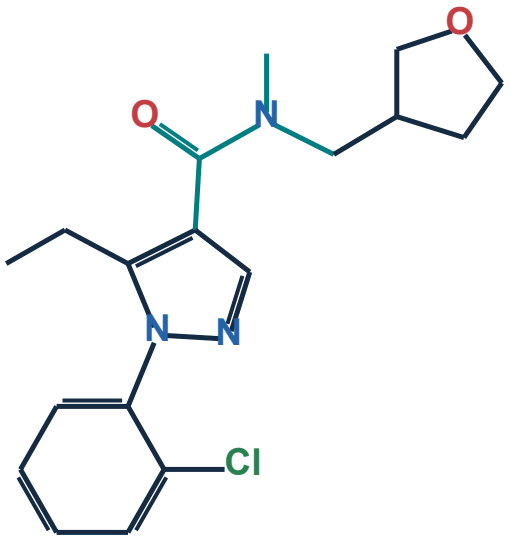
Our phase1 example

CCc1c(C(=O)N(C)CC2CCOC2)cnn1-c1ccccc1Cl

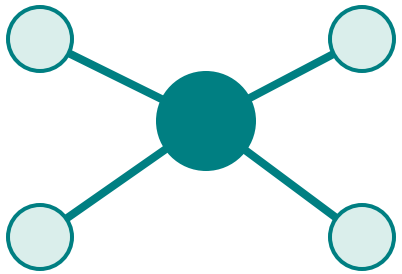
Chemical representations become model inputs

The local baseline uses Morgan fingerprints with 2,048 bits and radius 2

Molecular graph



Local atom environments



Neighbourhoods around each atom
contribute to the fingerprint.

Bit vector

0 1 0 0 1 1 0 …

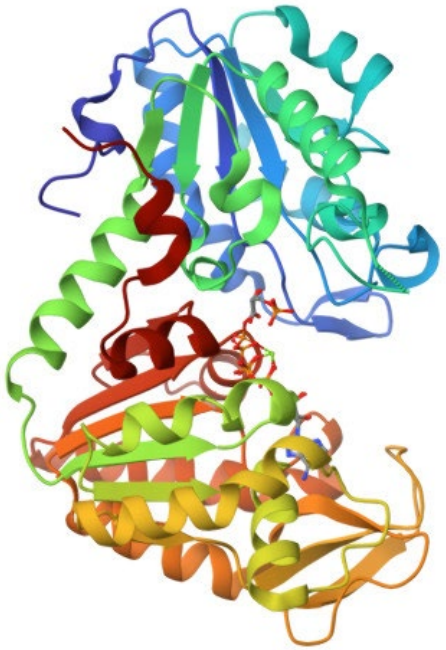
Schematic bits

Compression or feature selection
can prepare a quantum input.

The representation controls which chemical information the learner can use.

Protein–compound binding

A protein is a folded amino-acid chain with chemically distinct binding sites



Human PGK1, PDB 2X15

Structural example from your folder

Binding-site interactions (schematic)

Protein — Ligand

Hydrogen bond

N–H - - - - - - - - - - O=C

COO^- - - - - - - - - - ^+H_3N

Charge attraction

Shape and close contacts also matter.
Water and molecular flexibility change the free-energy balance.

P + L ⇌ PL

Phase 1: screening compounds for protein binding

ASMS identifies candidate binders through affinity-selection mass spectrometry

Phase 1: Affinity Selection MS

Protein + compound pool → **Separate bound fraction** → **Identify by mass**

Affinity-selection mass spectrometry detects compounds retained with the protein.

Binding, inhibition and affinity

A binding signal identifies a candidate. Follow-up assays can test effects on protein activity and quantify binding strength.

Local labels: 39 positives among 6,580 rows (0.59%).

The older DEL dataset was a development proxy. This presentation focuses on Phase 1.